

Qdrant Migration and Reranker Preview

KernelPack RAG — Experiment 03

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Codebase: kernelpack-python @ b0cc141 **Eval set:** qa_pairs/solvers_qa.json — 10 pairs, kernelpack.solvers, tiers: api (4), workflow (3), conceptual (3).

Baseline: v3 index rehydrated from chroma_export.pkl (solvers-only, LLM summaries prepended, UniXcoder embeddings, identifier-aware BM25+RRF).

Note: QA pairs are Codex-generated and not yet validated by a domain expert. Recall numbers are directional only.

Migration Verification

Every point in the Qdrant collection was verified against chroma_export.pkl: stable point ID present, payload text and metadata fields matching the Chroma export, and vector within 10^{-5} absolute tolerance. No mismatches. Count matched.

Recall Results

Backend parity: ChromaDB dense vs Qdrant dense

k	ChromaDB dense	Qdrant dense
3	3/10 (30%)	3/10 (30%)
5	5/10 (50%)	5/10 (50%)
10	8/10 (80%)	8/10 (80%)

Recall matches exactly. Ordered ID lists also match for every solver QA query at $k = 3, 5, 10$ — not just recall, but rank-for-rank agreement.

Hybrid parity: ChromaDB v3 hybrid vs Qdrant hybrid

k	ChromaDB v3 hybrid	Qdrant hybrid
3	8/10 (80%)	8/10 (80%)
5	10/10 (100%)	10/10 (100%)
10	10/10 (100%)	10/10 (100%)

Qdrant reproduces the 02b hybrid retrieval path when dense parity holds and BM25+RRF are mirrored. The 8/10 at recall@3 matches the 02b authoritative result.

Downstream experiment: Qdrant dense vs Qdrant reranked

k	Qdrant dense	Qdrant reranked
3	3/10 (30%)	8/10 (80%)
5	5/10 (50%)	8/10 (80%)
10	8/10 (80%)	10/10 (100%)

A cross-encoder reranker (`ms-marco-MiniLM-L-6-v2`, `fetch_k=30`) adds +5 `recall@3` over dense-only Qdrant. This is a downstream retrieval experiment, not migration evidence — it is reported separately because it changes the pipeline beyond storage.

Key Findings

1. **Migration is clean.** Point-by-point verification passed. Dense ordered-ID parity confirms Qdrant is a drop-in storage replacement for the 02b dense leg.
2. **Hybrid parity holds.** The 02b recall numbers are reproducible on Qdrant when BM25+RRF are mirrored. The migration does not degrade retrieval.
3. **The reranker result confirms the 02b diagnostic.** The 02b wide retrieval diagnostic showed all 10 targets in the top-50 dense window and `recall@5 = 10/10`, predicting that a reranker window of 5 would recover buried targets. The reranker result here — +5 at `recall@3` over dense-only — is consistent with that prediction. Dense-only `recall@3` (3/10) is the floor; hybrid `recall@3` (8/10) and reranked `recall@3` (8/10) are the same ceiling from different paths.

Open Issues

Metric alignment (P1). Qdrant uses cosine distance; ChromaDB commentary in 02b describes L2. Embeddings are not unit-normalized, so the two metrics can rank differently. Dense ordered-ID parity happens to hold on this eval set, but the metric equivalence is not proven. Explicitly pin both collections to the same metric space before expanding the eval set.

Notebook-local BM25 (P1). The Qdrant hybrid path in this notebook reuses the in-memory BM25 index from the rehydrated ChromaDB baseline. This is not a production Qdrant hybrid implementation — it is a parity check. Moving to Qdrant native sparse vectors is required before treating this as a deployed hybrid retriever.

Next Steps

1. **v4 — cross-encoder reranker.** Implement on the Qdrant hybrid pipeline (not dense-only). The 02b wide diagnostic gives `recall@5 = 10/10` under hybrid; a reranker over a hybrid candidate pool has a higher ceiling than reranking dense-only. Measure delta on the same 10-query eval.
2. **Qdrant native hybrid search.** Replace notebook-local BM25+RRF with Qdrant's native sparse + dense hybrid. Verify the same `k=3/5/10` ordered-ID behavior as the notebook-local path before treating the migration as complete.
3. **Resolve metric alignment.** Pin ChromaDB and Qdrant to the same distance metric and re-verify ordered-ID parity.
4. **Expand eval set.** Current eval covers `kernelpack.solvers` only (10 pairs). Expand to all 6 modules (target 50–100 pairs) and validate with a domain expert before drawing conclusions about the full pipeline.